Final Project
Part 4 - Migrating to Docker Swarm, Implementing Monitoring, and Preparing the Final Submission

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Virtualization and Cloud Computing - ay 2025/2026

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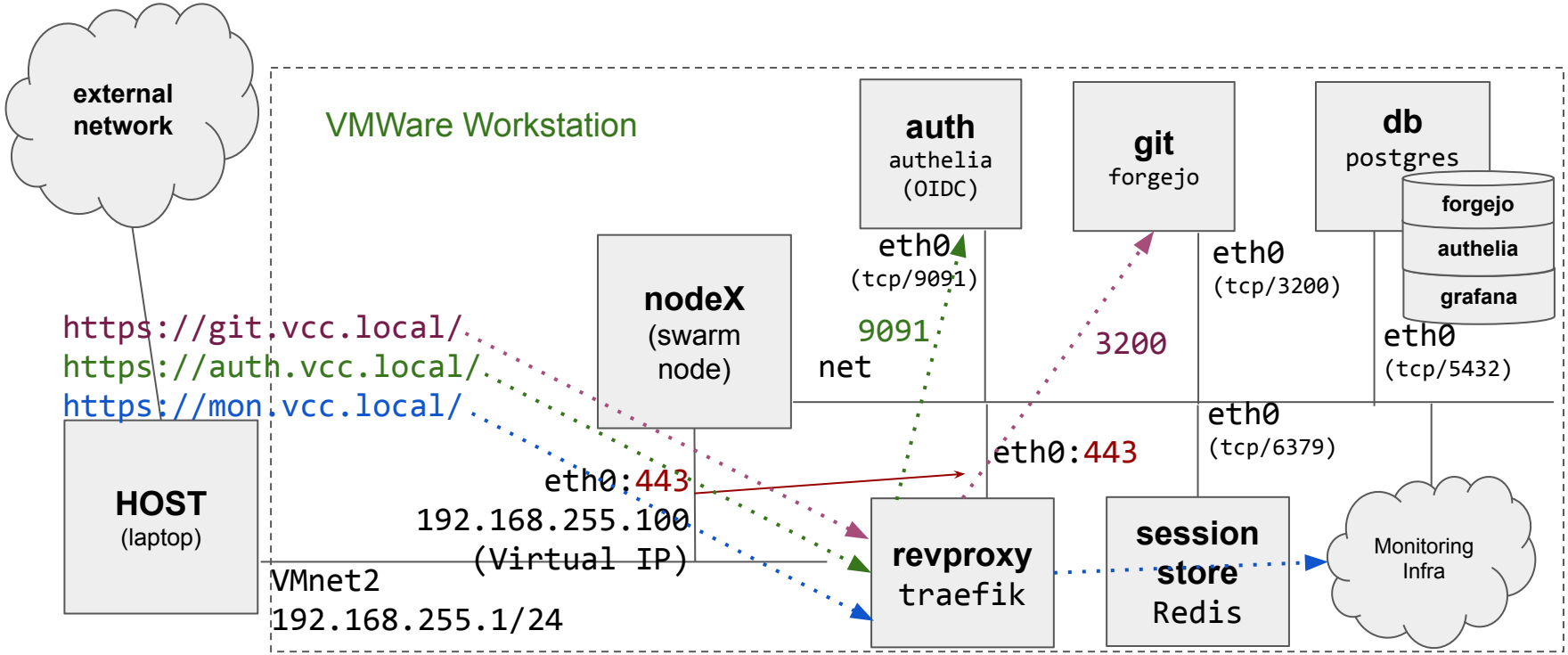
Objectives

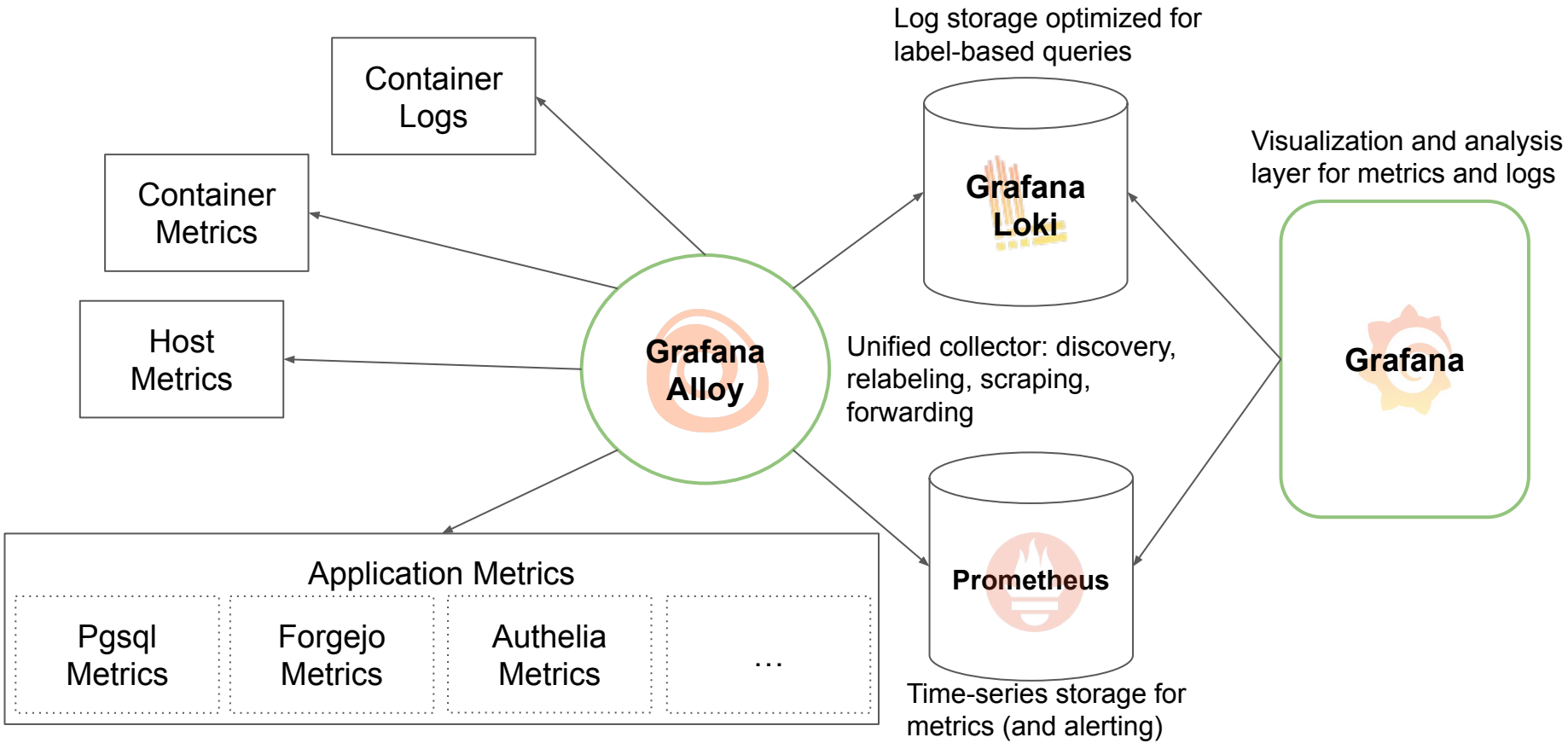
This last part focuses on migrating services from Docker Compose to Docker Swarm:

- Core services (Forgejo, Authelia, and Grafana) need to be configured for starting replicated and being scaled up/down within the Swarm cluster (a Redis backend is added to provide a shared session store for Authelia)
- Understand the monitoring infrastructure
- Extend SSO to Grafana
- Configure Alloy to scrape Authelia, Forgejo, Postgres, and Redis
- Create meaningful Grafana dashboards
- Protect with Vault all the passwords and secrets before submitting them to the repo (use `VCCsecret` as the secret to encrypt them)

Part4: layout



Monitoring infrastructure



Activities

- All required activities are detailed through the TODO items included in the project template
- These TODOs cover every part of the assignment (Parts 1, 2, 3, and 4) and are listed both in `TODO.md` and inside the project files
- The TODO Tree extension in VS Code can help to locate and track them quickly

Rules

- The final project can be completed individually or in groups of up to 3 members
- The project template is available on GitHub Classroom (classroom code: https://classroom.github.com/a/szBZZXKD)
- All members of a group must join the same Classroom assignment
- The repository created through Classroom is part of the official submission of the project
- This project is valid only for the 7 exam sessions of 2026 (no extensions, exceptions, or late submissions will be accepted)

Deliverables

- The completed project in your GitHub Classroom repository
- A short presentation of 6 slides, as below:
  1. Who You Are (group identification)
  2. Project Overview (your interpretation)
  3. What You Learned (technical concepts)
  4. The Most Interesting/Surprising Issue Encountered
  5. Key Metric + Log Insight (demo preview)*
  6. Possible Improvements

*You should use scripts that actively stimulate the system in order to generate the metrics and make the presented dashboard meaningful (e.g., use tools such as `curl`, `iptables`, `tc`, or controlled fault injections)

Exam

- Register for one of the following dates: 14/01/2026, 30/01/2026, 19/02/2026, 10/06/2026, 30/06/2026, 20/07/2026, 16/09/2026
- All group members must be registered for the same exam date, and must specify the name of their GitHub Classroom repository
- The oral exam consists of:
  - the 6-slide presentation
  - a live demo of the project
  - technical questions to all members about the project, including configuration changes and code explanations

The final grade is shared by all members of the group.